STANDARD 2

Stage 6

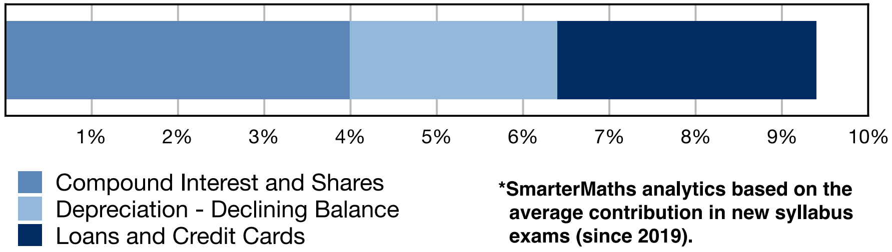
Financial Maths (Std 2), F4 Investments and Loans (Y12)

Loans and Credit Cards (Std 2)

Teacher: Hailey Summer

Exam Equivalent Time: 123 minutes *(based on allocation of 1.5 minutes per mark)*

STD2 Exam Contribution History
F4 - Investments and Loans



***SmarterMaths analytics based on the average contribution in new syllabus exams (since 2019).**

IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- *MS-F4 Investments and Loans* has contributed a substantial 9.4% to the new syllabus Std2 exams since its introduction in 2019.
- We have split this area into 3 sub-topics for analysis purposes: *Compound Interest and Shares (4.0%), Depreciation - Declining Balance (2.4%)* and *Loans and Credit Cards (3.0%).*
- This analysis looks at the largest sub-topic *Loans and Credit Cards (3.0%).*

ANALYSIS - What to Expect and Common pitfalls

- *Credit cards* were tested in a 4-mark question in 2023 after a two year absence, producing sub 50% mean marks in both parts..
- Calculating daily interest rates and applying the correct number of days has proven very challenging for the majority of students. Numerous database questions look at this area and deserve careful attention.
- A number of past HSC credit card questions have been adjusted from simple to daily compounding interest rates to align with the new syllabus (these have * in the title).
- Loan payment tables appeared in the 2023 and 2022 exams after a 3 year absence. The 2023 question caused carnage with mean marks of 50% and 11% in its two parts - a "must review" question!
- Effective revision in this area should provide exposure to a number of different styles of such tables (there are many variations).
- *"Loan P+I-R Table"* style questions were asked in 2022, 2017 and 2016, receiving significant mark allocations each time. Interpreting these tables caused significant problems on each occasion and revision attention is recommended.

Questions

1. Financial Maths, STD2 F4 SM-Bank 2 MC

Sally purchased an electronic game machine on hire purchase. She paid \$140 deposit and then \$25.50 per month for two years.

The total amount that Sally paid is

A. \$191

B. \$446

C. \$612

D. \$752

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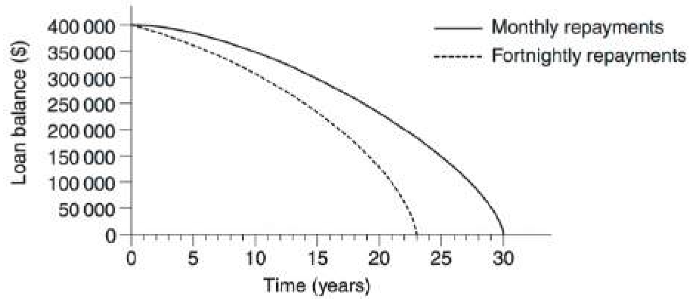
2. Financial Maths, STD2 F4 2012 HSC 24 MC

A \$400 000 loan can be repaid by making either monthly or fortnightly repayments.

The graph shows the loan balances over time using these two different methods of repayment.



The monthly repayment is \$2796.86 and the fortnightly repayment is \$1404.76.

What is the difference in the total interest paid using the two different methods of repayment, to the nearest dollar?

A. \$51 596

B. \$166 823

C. \$210 000

D. \$234 936

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3. Financial Maths, STD2 F4 2005 HSC 10 MC

The table is used to calculate monthly loan repayments.

Monthly loan repayments (in dollars) per $1000 borrowed

Interest rate % pa	*5 years*	*10 years*	*15 years*	*20 years*
5%	18.87	10.61	7.91	6.60
6%	19.33	11.10	8.44	7.16
7%	19.80	11.61	8.99	7.75
8%	20.28	12.13	9.56	8.36
9%	20.76	12.67	10.14	9.00

Samantha has borrowed $70 000 at 8% per annum for 15 years.

What is her monthly loan repayment?

A. $143.40

B. $669.20

C. $8030.40

D. $10 038.00

4. Financial Maths, STD2 F4 2008 HSC 15 MC

Ali is buying a speedboat at Betty's Boats.

Betty's Boats

Cash price
$16 000

OR

Terms
15% deposit plus
$320 per month for
5 years

What is the amount of interest Ali will have to pay if he chooses to buy the boat on terms?

A. $3200

B. $5600

C. $19 200

D. $21 600

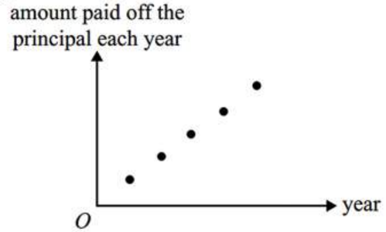
5. Financial Maths, STD2 F4 SM-Bank 1 MC

Ernie took out a reducing balance loan to buy a new family home.

He correctly graphed the amount **paid off** the principal of his loan each year for the first five years.

The shape of this graph (for the first five years of the loan) is best represented by

A.



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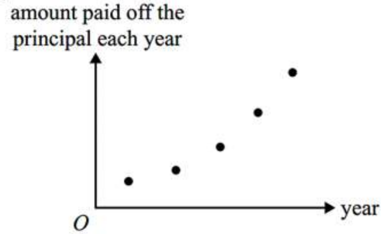

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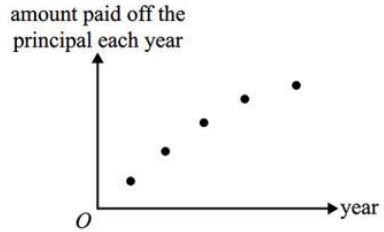

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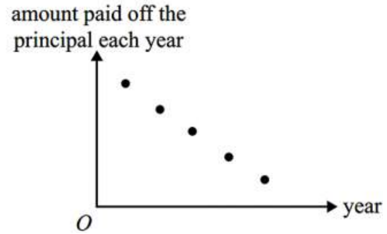

6. Financial Maths, STD2 F4 2011* HSC 10 MC

A television was purchased for $2100 on 12 April 2011 using a credit card. Compound interest was charged daily at a rate basis 19.71% per annum for purchases on this credit card. There were no other purchases on this credit card account.

There was no interest-free period. The period for which interest was charged included the date of purchase and the date of payment.

What amount was paid when the account was paid in full on 20 May 2011?

A. $2143.09

B. $2143.53

C. $2144.23

D. $2144.68

7. Financial Maths, STD2 F4 2011 HSC 22 MC

Ying borrowed $250 000 to buy a house. The interest rate and monthly repayment for her loan are shown in the spreadsheet.

	A	B	C	D	E
1	Home Loan Table			This table assumes the same number of days in each month, ie	
2		Amount	= $250 000		
3		Annual Interest Rate	= 7.65%	Interest = Rate/12 × Principal	
4		Monthly Repayment (R)	= $1871.94		
5					
6	Month	Principal (P)	Interest (I)	$P + I$	$P + I - R$
7	1	$250 000.00	$1593.75	$251 593.75	$249 721.81
8	2	$249 721.81	$1591.98	$251 313.79	$249 441.85
9	3	$249 441.85	$1590.19	$251 032.04	
10	4				

Sheet 1 Sheet 2

What is the total interest charged for the first four months of this loan?

A. $6364.32

B. $6366.11

C. $6369.67

D. $6376.25

8. Financial Maths, STD2 F4 2004 HSC 19 MC

Kerry has a credit card. She is charged 0.05% compound interest per day on outstanding balances.

How much interest is Kerry charged on an amount of $250, which is outstanding on her credit card for 30 days?

A. $3.75

B. $3.78

C. $253.75

D. $253.78

9. Financial Maths, STD2 F4 2006 HSC 21 MC

Bill borrows $420 000 to buy a house. Interest is charged at 7.2% per annum, compounded monthly.

How much does he owe at the end of the first month, after he has made a $4000 repayment?

A. $418 496

B. $418 520

C. $445 952

D. $446 240

10. Financial Maths, STD2 F4 2016 HSC 17 MC

Ariana is charged compound interest at the rate of 0.036% per day on outstanding credit card balances. She has $780 outstanding for 24 days.

How much compound interest is she charged?

A. $6.74

B. $6.77

C. $786.74

D. $786.77

11. Financial Maths, STD2 F4 2010* HSC 22 MC

In July, Ms Alott received a statement for her credit card account. The account has no interest free period. Compound interest is calculated daily and charged to her account on the statement date.

Ms I O Alott

Dollar Bank

Credit limit: \$5000

Credit card statement

Statement date: 23 July 2010

Previous Balance	Payments	Purchases	Interest charged
\$529.46	\$529.46	\$1721.50	?

Date	Purchases	Amount	Closing Balance
29 June	Hi-Fi	\$1721.50	?

Annual percentage rate: 19%

Daily percentage rate: 0.0521%

Note: Interest is charged on amounts from (and including) the date of purchase up to (and including) the statement date.

Minimum payment due: \$25 or 5% of closing balance whichever is the larger.

What is the minimum payment due on this account?

A. \$23.56

B. \$25.00

C. \$86.08

D. \$87.20

12. Financial Maths, STD2 F4 2009 HSC 26c

Margaret borrowed \$300 000 to buy an apartment. The interest rate is 6% per annum, compounded monthly. The repayments were set by the bank at \$2200 per month for 20 years.

The loan balance sheet shows the interest charged and the balance owing for the first month.

Month	*Principal at the start of the month*	*Monthly interest*	*Monthly repayment*	*Balance at end of month*
1	\$300 000	\$1500	\$2200	\$299 300
2	\$299 300	***A***	\$2200	***B***

i. What is the total amount that is to be paid for this loan over the 20 years? ***(1 mark)***

ii. Find the values of A and B. ***(2 marks)***

13. Financial Maths, STD2 F4 2018 HSC 29e

Andrew borrowed \$20 000 to be repaid in equal monthly repayments of \$243 over 10 years. Having made this monthly repayment for 4 years, he increased his monthly repayment to \$281. As a result, Andrew paid off the loan one year earlier.

How much less did he repay altogether by making this change? ***(2 marks)***

14. Financial Maths, STD2 F4 2020 HSC 22

Nisa has a credit card on which interest at 17% per annum, compounded daily, is charged on the amount owing.

At the beginning of the month, Nisa owes \$500 on her credit card. She makes no other purchases using the credit card, but fifteen days later, she repays \$250.

Assuming that interest is charged for the fifteen days, calculate the amount owing on the credit card immediately after the \$250 payment is made. ***(3 marks)***

15. Financial Maths, STD2 F4 2024 HSC 27

A couple borrows \$30 000 to be repaid in equal monthly repayments of \$280 over 10 years.

After following this repayment plan for 5 years, they decide to decrease their monthly repayment to \$250. As a result, it will take them an additional two years to pay off their loan.

How much more will they repay in total by making this change? ***(3 marks)***

16. Financial Maths, STD2 F4 2016 HSC 27d

Marge borrowed \$19 000 to buy a used car. Interest on the loan was charged at 4.8% pa at the end of each month. She made a repayment of \$436 at the end of every month. The table below sets out her monthly repayment schedule for the first four months of the loan.

Month	*Amount owing at start of month*	*Interest charged*	*Repayment*	*Amount owing at end of month*
1	A	\$76.00	\$436.00	\$18 640.00
2	\$18 640.00	X	\$436.00	\$18 278.56
3	\$18 278.56	\$73.11	\$436.00	\$17 915.67
4	\$17 915.67	\$71.66	\$436.00	B

i. Some values in the table are missing. Write down the values for A and B. ***(2 marks)***

ii. Calculate the value of X. ***(2 marks)***

iii. Marge repaid this loan over four years.
What is the total amount that Marge repaid? ***(1 mark)***

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17. Financial Maths, STD2 F4 2004 HSC 27a

Aaron decides to borrow \$150 000 over a period of 20 years at a rate of 7.0% per annum.

MONTHLY REPAYMENT TABLE

Principal and interest per \$1000 borrowed

Interest rate (pa)	**Term of loan – years**					
	5	**10**	**15**	**20**	**25**	**30**
6.5%	19.57	11.35	8.71	7.46	6.75	6.32
7.0%	19.80	11.61	8.99	7.75	7.07	6.65
7.5%	20.04	11.87	9.27	8.06	7.39	6.99
8.0%	20.28	12.13	9.56	8.36	7.72	7.34

i. Using the Monthly Repayment Table, calculate Aaron's monthly repayment. ***(2 marks)***

ii. How much interest does he pay over the 20 years? ***(2 marks)***

iii. Aaron calculates that if he repays the loan over 15 years, his total repayments would be \$242 730.
How much interest would he save by repaying the loan over 15 years instead of 20 years? ***(2 marks)***

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18. Financial Maths, STD2 F4 2006* HSC 25b

In June, Ms Bigspender received a statement for her credit card account.

The account has no interest-free period. Compound interest is calculated daily and charged to her account on the statement date.

Ms Ima Bigspender
Credit limit: \$2000

Statement date: 20 June 2006

Sum Bank
Credit Card Statement

Previous balance	**Payments**	**Purchases**	**Interest charged**
\$263.83	**\$263.83**	**\$617.72**	

Date	*Purchases*	*Amount*
23 May	Concert tickets	\$617.72

Annual percentage rate: 18.2%
Daily percentage rate: 0.0498%

Note: Interest is charged on amounts from (and including) the date of purchase up to (and including) the statement date.

i. For how many days is she charged interest on her purchase? ***(1 mark)***

ii. Calculate the interest charged to her account. ***(2 marks)***

19. Financial Maths, STD2 F4 2006 HSC 27a

Liliana wants to borrow money to buy a house. The bank sent her an email with the following table attached.

Monthly repayments

Amount borrowed	Term of loan				
	10 years 120 months	15 years 180 months	20 years 240 months	25 years 300 months	30 years 360 months
$80 000	$970.62	$764.52	$669.15	$617.45	$587.01
$90 000	$1091.95	$860.09	$752.80	$694.63	$660.39
$100 000	$1213.28	$955.65	$836.44	$771.82	$733.76
$110 000	$1334.60	$1051.22	$920.08	$849.00	$807.14
$120 000	$1455.93	$1146.78	$1003.73	$926.18	$880.52
$130 000	$1577.26	$1242.35	$1087.37	$1003.36	$953.89
$140 000	$1698.59	$1337.91	$1171.02	$1080.54	$1027.27
$150 000	$1819.91	$1433.48	$1254.66	$1157.72	$1100.65
$160 000	$1941.24	$1529.04	$1338.30	$1234.91	$1174.02

i. Liliana decides that she can afford $1000 per month on repayments.
What is the maximum amount she can borrow, and how many years will she have to repay the loan? ***(1 mark)***

ii. Zali intends to borrow $160 000 over 15 years from the same bank.
If she chooses to borrow $160 000 over 20 years instead, how much more interest will she pay? ***(2 marks)***

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20. Financial Maths, STD1 F3 2021 HSC 30

Blake opens a new credit card account on 1 May. He uses it, for the first time, on 4 May to buy concert tickets for $850.

He makes no further purchases or repayments during the month of May.

A statement for the credit card is issued on the last day of each month.

The statement for May shows that interest is charged at 19.75% per annum, compounding daily, from 20 May (included) until 31 May (included).

a. What is the compound interest shown on the statement issued on 31 May? ***(3 marks)***

b. The minimum payment is calculated as 3% of the closing balance on 31 May. Calculate the minimum payment. ***(1 mark)***

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21. Financial Maths, STD2 F4 2024 HSC 21

William has a reducing balance loan on which he owes $5590. He makes monthly repayments of $110.

The loan company charges interest at 24% per annum, compounded monthly.

The spreadsheet shows some of the information for the next two months of the loan.

a. Complete the entries in the spreadsheet to show the balance owing on the loan at the end of two months. ***(2 marks)***

	A	B	C	D	E
1	Month	Principal	Interest charged	Amount repaid	Balance owing
2	1	$5590.00	$111.80	$110.00	
3	2			$110.00	

b. Explain why the loan will never be repaid if William continues to make repayments of $110 per month. ***(1 mark)***

22. Financial Maths, STD2 F4 2010 HSC 28a

The table shows monthly home loan repayments with interest rate changes from February to October 2009.

Monthly home loan repayments

	A	B	C	D	E
1					
2	*Dates*	Feb 2009	Apr 2009	Jun 2009	Oct 2009
3	*Increase/Decrease*	*−1.0%*	*−0.1%*	*0.05%*	*0.25%*
4	***Rate***	***5.85%***	***5.75%***	***5.80%***	***6.05%***
5	$1 000	$6.35	$6.29	$6.32	$6.47
6	$50 000	$318	$315	$316	$324
7	$100 000	$635	$629	$632	$647
8	$150 000	$953	$944	$948	$971
9	$200 000	$1 270	$1 258	$1 264	$1 295
10	$250 000	$1 588	$1 573	$1 580	$1 618
11	$300 000	$1 905	$1 887	$1 896	$1 942
12	$350 000	$2 223	$2 202	$2 212	$2 266
13	$400 000	$2 541	$2 516	$2 529	$2 589
14					

Sheet 1 Sheet 2

i. What is the change in monthly repayments on a $250 000 loan from February 2009 to April 2009? ***(1 mark)***

ii. Xiang wants to borrow $307 000 to buy a house.

Xiang's bank approves loans for customers if their loan repayments are no more than 30% of their monthly gross salary.

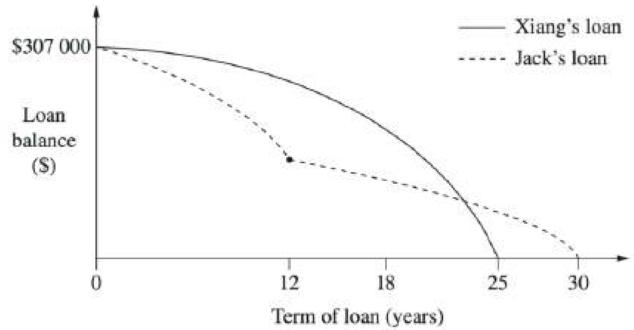
Xiang's monthly gross salary is $6500.

If she had applied for the loan in October 2009, would her bank have approved her loan?

Justify your answer with suitable calculations. ***(3 marks)***

iii. Jack took out a loan at the same time and for the same amount as Xiang.

Graphs of their loan balances are shown.



Identify TWO differences between the graphs and provide a possible explanation for each difference, making reference to interest rates and/or loan repayments. ***(2 marks)***

23. Financial Maths, STD2 F4 2010 HSC 25b

William wants to buy a car. He takes out a loan for $28 000 at 7% per annum interest for four years.

Monthly repayments for loans at different interest rates are shown in the spreadsheet.

	A	B	C	D	E
1		**Monthly repayments**			
2		**Term of loan (in months)**		48	
3					
4	**Amount**	**Interest rate p.a.**			
5	**borrowed**	6%	7%	8%	9%
6	$27 000	$634.10	$646.55	$659.15	$671.90
7	$27 500	$645.84	$658.52	$671.36	$684.34
8	$28 000	$657.58	$670.49	$683.56	$696.78
9	$28 500	$669.32	$682.47	$695.77	$709.22
10	$29 000	$681.07	$694.44	$707.97	$721.67
11	$29 500	$692.81	$706.41	$720.18	$734.11
12	$30 000	$704.55	$718.39	$732.39	$746.55

Sheet 1 Sheet 2

How much interest does William pay over the term of this loan? ***(2 marks)***

24. Financial Maths, STD2 F4 2012* HSC 26c

Heather used her credit card to purchase a plane ticket valued at $1990 on 28 January 2011. She made no other purchases on her credit card account in January. She paid the January account in full on 19 February 2011.

The credit card account has no interest free period. Compound interest is charged daily at the rate of 21.9% per annum, including the date of purchase and the date the account is paid.

How much interest did she pay, to the nearest cent? ***(2 marks)***

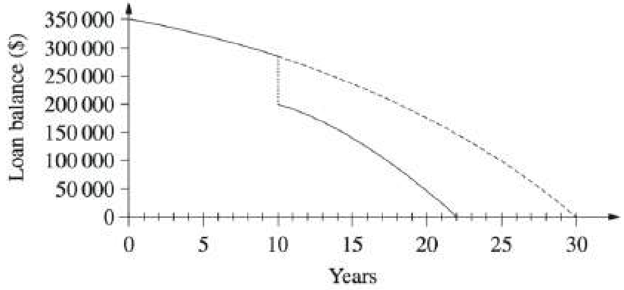
25. Financial Maths, STD2 F4 2015* HSC 29a

On 20 August, tickets were purchased for $425 using a credit card. No other purchases were made using this card in August. Compound interest was charged daily at a rate of 18.25% per annum. There was no interest-free period. The period for which interest was charged included the date of purchase and the date of payment.

What amount was paid when the account was paid in full on 31 August? ***(2 marks)***

26. Financial Maths, STD2 F4 2015 HSC 29b

Jamal borrowed $350 000 to be repaid over 30 years, with monthly repayments of $1880. However, after 10 years he made a lump sum payment of $80 000. The monthly repayment remained unchanged. The graph shows the balances owing over the period of the loan.



Over the period of the loan, how much less did Jamal pay by making the lump sum payment? ***(2 marks)***

27. Financial Maths, STD2 F4 2017 HSC 28c

Michelle borrows $100 000. The interest rate charged is 12% per annum compounded monthly. The monthly payment is $1029 and the first repayment is made after one month.

What is the amount outstanding immediately after the SECOND monthly repayment is made? ***(3 marks)***

28. Financial Maths, STD2 F4 2018* HSC 28d

Yanika opens a new credit card account, with interest and fees as shown.

Interest

- Compound interest calculated daily at the rate 12.41% p.a.
- No interest-free period

Fees

- $0 for online repayments
- $3 for repayments in cash (fee added to balance immediately after repayment)

Yanika makes a single purchase of $849 with the credit card.

i. Show that the balance owing on the credit card 24 days after making the purchase is $855.95 . ***(2 marks)***

ii. Yanika makes her first repayment 24 days after making the purchase. She makes a cash repayment of $450.

What is the balance owing on the credit card immediately after her repayment is made and the repayment fee has been charged? ***(1 mark)***

29. Financial Maths, STD2 F4 2019 HSC 27

Ashley has a credit card with the following conditions:

- There is no interest-free period.
- Interest is charged at the end of each month at 18.25% per annum, compounding daily, from the purchase date (included) to the last day of the month (included).

Ashley's credit card statement for April is shown, with some figures missing.

1 April to 30 April		
Date	**Details**	**Amount ($)**
1 April	Opening balance	0
20 April	Furniture	3700
30 April	Interest charged	***
30 April	Closing balance	***

Minimum payment: ☐

The minimum payment is calculated as 2% of the closing balance on 30 April.

Calculate the minimum payment. ***(3 marks)***

30. Financial Maths, STD2 F4 2023 HSC 32

Ali has a credit card which has no interest-free period. Interest is charged at 13.5% per annum, compounding daily, on the amount owing.

During the month, Ali made only one purchase of $450 using the credit card. The full amount owing was repaid 21 days later.

a. Calculate the amount of interest charged on the purchase, assuming that interest is charged for the 21 days. ***(2 marks)***

b. What percentage of the full amount repaid is the interest? Give the answer to two decimal places. ***(2 marks)***

31. Financial Maths, STD2 F4 2022 HSC 36

Frankie borrows $200 000 from a bank. The loan is to be repaid over 23 years at a rate of 7.2% per annum, compounded monthly. The repayments have been set at $1485 per month.

The interest charged and the balance owing for the first three months of the loan are shown in the spreadsheet below.

Month	*Principal (at start of month)*	*Interest charged*	*Monthly repayment*	*Balance (at end of month)*
1	$200 000	$1200	$1485	$199 715
2	$199 715	***A***	$1485	$199 428.29
3	$199 428.29	$1196.57	$1485	***B***

a. What are the values of A and B? ***(2 marks)***

b. After 50 months of repaying the loan, Frankie decides to make a lump sum payment of $ 40 000 and to continue making the monthly repayments of $1485. The loan will then be fully repaid after a further 146 monthly repayments.

How much less will Frankie pay overall by making the lump sum payment? ***(3 marks)***

32. Financial Maths, STD2 F4 2023 HSC 29

The table shows monthly repayments for each $1000 borrowed.

Monthly repayment table

Principal and Interest per $1000 borrowed

Interest rate (per annum)	*Term of loan* (years)					
	5	10	15	20	25	30
6.5%	19.57	11.35	8.71	7.46	6.75	6.32
7.0%	19.80	11.61	8.99	7.75	7.07	6.65
7.5%	20.04	11.87	9.27	8.06	7.39	6.99
8.0%	20.28	12.13	9.56	8.36	7.72	7.34

a. A couple borrows $520 000 to buy a house at 8% per annum over 25 years.

How much does the couple repay in total for this loan? ***(3 marks)***

b. Chris borrows some money at 7% per annum. Chris will repay the loan over 15 years, paying $3596 per month.

How much money does Chris borrow? ***(1 mark)***

Worked Solutions

1. Financial Maths, STD2 F4 SM-Bank 2 MC

$$\begin{aligned}\text{Total paid} &= 140 + 25.50 \times 2 \times 12\\ &= \$752\end{aligned}$$

$\Rightarrow D$

2. Financial Maths, STD2 F4 2012 HSC 24 MC

$$\begin{aligned}\text{Monthly repayment} &= \$2796.86\\ \#\,\text{Repayments} &= 30 \times 12 = 360\\ \text{Total repaid} &= 360 \times 2796.86\\ &= \$1\ 006\ 869.60\\ \text{Total interest} &= 1\ 006\ 869.60\ - 400\ 000\\ &= \$606\ 869.60\end{aligned}$$

$$\begin{aligned}\text{Fortnightly payment} &= \$1404.76\\ \#\,\text{Repayments} &= 23 \times 26 = 598\\ \text{Total repaid} &= 598 \times 1404.76\\ &= \$840\ 046.48\\ \text{Total interest} &= 840\ 046.48 - 400\ 000\\ &= \$440\ 046.48\end{aligned}$$

$$\begin{aligned}\therefore\ \text{Difference in interest} &= 606\ 869.60 - 440\ 046.48\\ &= \$166\ 823 \quad \text{(nearest dollar)}\end{aligned}$$

$\Rightarrow\ B$

3. Financial Maths, STD2 F4 2005 HSC 10 MC

Monthly repayment of \$1000 at 8% for 15 years

$= \$9.56$

$\therefore$ Monthly repayment of \$70 000

$= 70 \times \$9.56$

$= \$669.20$

$\Rightarrow\ B$

4. Financial Maths, STD2 F4 2008 HSC 15 MC

$$\begin{aligned}\text{Deposit} &= 15\% \times 16\ 000\\ &= 2400\end{aligned}$$

$$\begin{aligned}\text{Payments} &= 320 \times 5 \times 12\\ &= 19\ 200\end{aligned}$$

$$\begin{aligned}\text{Total paid} &= 2400 + 19\ 200\\ &= 21\ 600\end{aligned}$$

$$\begin{aligned}\therefore\ \text{Interest} &= 21\ 600 – 16\ 000\\ &= 5600\end{aligned}$$

$\Rightarrow\ B$

5. Financial Maths, STD2 F4 SM-Bank 1 MC

A reducing balance loan means that the amount of interest paid out decreases each year, and therefore the amount paid off the principal will not only increase each year, but will do so at an increasing rate.

B correctly shows this trend.

$\Rightarrow\ B$

6. Financial Maths, STD2 F4 2011* HSC 10 MC

$$\begin{aligned}\text{Days for interest }(n) &= 19 + 20 = 39\\ \text{Daily interest rate }(r) &= \frac{0.1971}{365} = 0.00054\\ \text{Total paid }(FV) &= PV(1+r)^n\\ &= 2100(1.00054)^{39}\\ &= \$2144.68\end{aligned}$$

$\Rightarrow D$

♦♦ *Mean mark 32%*
COMMENT: Make sure you know how to adjust an annual rate to a daily rate, as shown in the Worked Solutions.

7. Financial Maths, STD2 F4 2011 HSC 22 MC

Month 3

$P + I - R = 251\ 032.04 - 1871.94$

$= 249\ 160.10$

Month 4

$P = 249\ 160.10$

$\therefore I = 249\ 160.10 \times \dfrac{0.0765}{12}$

$= 1588.40$

$\therefore$ Total interest $= 1593.75 + 1591.98 + 1590.19 + 1588.40$

$= 6364.32$

$\Rightarrow A$

---

8. Financial Maths, STD2 F4 2004 HSC 19 MC

Using $A = P(1+r)^n$

$A = 250(1 + 0.0005)^{30}$

$= 250(1.0151\ldots)$

$= 253.777\ldots$

$\therefore$ Amount of interest charged

$= 253.777\ldots - 250$

$= \$3.777\ldots$

$\Rightarrow B$

---

9. Financial Maths, STD2 F4 2006 HSC 21 MC

Let L = Amount of the loan after 1 month

$r = \dfrac{7.2\%}{12} = 0.6\% = 0.006$

Using $FV = PV(1+r)^n$

$L = 420\ 000(1 + 0.006)^1 - \text{repayment}$

$= 420\ 000(1.006) - 4000$

$= \$418\ 520$

$\Rightarrow B$

---

10. Financial Maths, STD2 F4 2016 HSC 17 MC

Total owing

$= P(1+r)^n$

$= 780\left(1 + \dfrac{0.036}{100}\right)^{24}$

$= 786.77$

♦ *Mean mark 38%.*
COMMENT: Credit card problems consistently produce sub-50% mean marks. Important review area.

$\therefore$ Interest charged

$= 786.77 - 780$

$= \$6.77$

$\Rightarrow B$

---

11. Financial Maths, STD2 F4 2010* HSC 22 MC

Days of interest $(n) = 2 + 23 = 25$

Daily interest rate $(r) = \dfrac{0.0521}{100} = 0.000521$

♦♦♦ *Mean mark 20%. Lowest scoring MC question in the 2010 exam.*

Closing Balance $(FV) = PV(1+r)^n$

$= 1721.50(1.000521)^{25}$

$= \$1744.06$

5% of closing balance $= \dfrac{5}{100} \times 1744.06$

$= 87.20$ (nearest cent)

Since $\$87.20 > \$25$, minimum payment $= \$87.20$

$\Rightarrow D$

---

12. Financial Maths, STD2 F4 2009 HSC 26c

i. Monthly repayment $= \$2200$

$\#$ Repayments $= 20 \times 12 = 240$

$\therefore$ Total paid $= 2200 \times 240$

$= \$528\ 000$

♦ *Mean mark 39%*
MARKER'S COMMENT: 1 mark allocation flags the answer should not be too involved.

ii. Interest rate monthly $= \dfrac{6\%}{12} = 0.5\%$

$A =$ Principal at start of month $\times \dfrac{0.5}{100}$

$= 299\ 300 \times \dfrac{0.5}{100}$

$= \$1496.50$

$B =$ Principal + interest – repayment

$= 299\ 300 + 1496.50 - 2200$

$= \$298\ 596.50$

---

13. Financial Maths, STD2 F4 2018 HSC 29e

Total original repayments $= 10 \times 12 \times 243$

$= \$29\ 160$

Actual repayments $= 4 \times 12 \times 243\ +\ 5 \times 12 \times 281$

$= \$28\ 524$

$\therefore$ Savings $= 29\ 160–28\ 524$

$= \$636$

---

14. Financial Maths, STD2 F4 2020 HSC 22

Days of interest $(n) = 15$

Daily interest rate $(r) = \dfrac{0.17}{365} = 0.00046575\ldots$

Amount owing $(FV) = PV(1+r)^n – 250$

$= 500(1.00046575\ldots)^{15} – 250$

$= 503.50 – 250$

$= \$253.50$

---

15. Financial Maths, STD2 F4 2024 HSC 27

Original plan:

Total repayments $= 10 \times 12 = 120$

Total repayments $= 120 \times 280 = \$33\ 600$

New plan:

Repayments at \$280 $= 5 \times 12 = 60$

Repayments at \$250 $= 7 \times 12 = 84$

Total repayments $= 60 \times 280 + 84 \times 250 = \$37\ 800$

Extra repaid $= 37\ 800 - 33\ 600 = \$4200$

---

16. Financial Maths, STD2 F4 2016 HSC 27d

i. $A + 76 - 436 = 18\ 640$

$\therefore A = \$19\ 000$

$17\ 915.67 + 71.66 - 436 = B$

$\therefore B = \$17\ 551.33$

ii. $18\ 640 + X - 436 = 18\ 278.56$

$\therefore X = 18\ 278.56 + 436 - 18\ 640$

$= \$74.56$

iii. Total amount repaid

$= 48 \times 436$

$= \$20\ 928$

♦♦ *Mean mark part (iii) 28%.*
COMMENT: Read carefully whether total paid or total *interest* paid is required.

---

17. Financial Maths, STD2 F4 2004 HSC 27a

i. Using the table:

Monthly repayment on \$1000 at 7.0% over 20 years = \$7.75

$\therefore$ Monthly repayment on \$150 000 loan

$= 150 \times 7.75$

$= \$1162.50$

ii. Total repayments over 20 years

$= 20 \times 12 \times 1162.50$

$= \$279\ 000$

$\therefore$ Interest paid over 20 years

$= 279\ 000 - 150\ 000$

$= \$129\ 000$

iii. Savings = Total paid (20 years) – Total paid (15 years)

$= 279\ 000 - 242\ 730$

$= \$36\ 270$

---

18. Financial Maths, STD2 F4 2006* HSC 25b

i. # Days interest charged

$= 9$ (in May) + 20 (in June)

$= 29$

ii. Daily interest rate $(r) = \dfrac{0.0498}{100} = 0.000498$

Closing Balance $(FV) = PV(1+r)^n$

$= 617.72(1.000498)^{29}$

$= \$626.703$

$\therefore$ Interest charged $= 626.70 - 617.72$

$= \$8.98$

---

19. Financial Maths, STD2 F4 2006 HSC 27a

i. From table

\$130 000 (over 30 years)

ii. Total repayments over 15 years

$= \$1529.04 \times 180$

$= \$275\ 227.20$

Total repayments over 20 years

$= \$1338.30 \times 240$

$= \$321\ 192.00$

$\therefore$ Extra interest over 20 years

$= 321\ 192.00 - 275\ 227.20$

$= \$45\ 964.80$

---

20. Financial Maths, STD1 F3 2021 HSC 30

a. Daily interest rate $= \dfrac{19.75}{365} = 0.05411\% = 0.0005411$

Days incurring interest = 12

Card balance (31 May) $= PV(1+r)^n$

$= 850(1+0.0005411)^{12}$

$= \$855.54$

$\therefore$ Interest $= 855.54 - 850$

$= \$5.54$

b. Minimum payment $= 855.54 \times 0.03$

$= \$25.67$

---

21. Financial Maths, STD2 F4 2024 HSC 21

a.

	A	B	C	D	E
1	Month	Principal	Interest charged	Amount repaid	Balance owing
2	1	\$5590.00	\$111.80	\$110.00	\$5591.80
3	2	\$5591.80	\$111.84	\$110.00	\$5593.64

Calculations:

Cell E2 $= 5590 + 111.80 - 110 = \$5591.80$

Cell B3 = Cell E2

Cell C3 $= 5591.80 \times 0.02 = \$111.84$ (monthly r/i = 2%)

Cell E3 $= 5591.80 + 111.84 - 110 = \$5593.64$

b. The interest charged exceeds the amount that is repaid.

Interest charges will gradually increase with the monthly repayment staying the same ⇒ loan will nev

22. Financial Maths, STD2 F4 2010 HSC 28a

i. Repayment (Feb 09) $= 1588$

Repayment (Apr 09) $= 1573$

Difference $= 1588 - 1573 = 15$

$\therefore$ Monthly repayments decrease by \$15

ii. Loan $= \$307\ 000$

$$\begin{aligned}\text{Repayments (Oct 09)} &= 1942 + (7 \times 6.47)\\ &= 1942 + 45.29\\ &= \$1987.29 \text{ per month}\end{aligned}$$

♦ *Mean mark 39%*
MARKER'S COMMENT: Borrowing \$307,000 can be achieved by borrowing \$300,000, and then **7** times the table repayment value for borrowing \$1000.

$$\begin{aligned}30\% \text{ Gross salary} &= 6500 \times 30\%\\ &= \$1950 \text{ per month}\end{aligned}$$

$\therefore$ Since repayments of \$1987.29 > \$1950, the loan would not have been approved.

iii. Differences

Jack's loan balance falls more sharply for first 12 years

Jack's loan balance falls less sharply between years 12-30.

Explanation(s)

Jack made larger repayments for 1st 12 years, or

Jack made the same repayments but had a lower interest rate for the first 12 years.

Jack made smaller repayments in years 12-30.

♦ *Mean mark 36%*
MARKER'S COMMENT: Explanations were generally poor and many failed to refer directly to the graphs shown, or reference Xiang or Jack directly.

23. Financial Maths, STD2 F4 2010 HSC 25b

Loan $= \$28\ 000$, $\quad r = 7\%$ p.a.

Monthly repayment $= \$670.49$

\# Repayments $= 4 \times 12 = 48$

$$\begin{aligned}\text{Total repaid} &= 48 \times 670.49\\ &= \$32\ 183.52\end{aligned}$$

$$\begin{aligned}\therefore \text{Interest paid} &= 32\ 183.52 - 28\ 000\\ &= \$4183.52\end{aligned}$$

♦ *Mean mark 42%*
MARKER'S COMMENT: An incorrect table value used correctly in the following calculations received half-marks here. *Show your working!*

24. Financial Maths, STD2 F4 2012* HSC 26c

$\text{Days of interest } (n) = 4 + 19 = 23$

$\text{Daily interest rate } (r) = \dfrac{0.219}{365} = 0.0006$

$$\begin{aligned}\text{Total Paid } (FV) &= PV(1+r)^n\\ &= 1990(1.0006)^{23}\\ &= 2017.644\ldots\\ &= \$2017.64\end{aligned}$$

$$\begin{aligned}\therefore \text{ Interest Paid} &= 2017.64 - 1990\\ &= \$27.64\end{aligned}$$

♦ Mean mark 38%
MARKER'S COMMENT: Too many students calculated the # days as 22 instead of 23!

---

25. Financial Maths, STD2 F4 2015* HSC 29a

$\text{Days of interest } = 12$

$\text{Daily interest rate } = \dfrac{0.1825}{365} = 0.0005$

$$\begin{aligned}\text{Total Paid } (FV) &= PV(1+r)^n\\ &= 425(1.0005)^{12}\\ &= 427.557\ldots\\ &= \$427.56 \ \ \text{(nearest cent)}\end{aligned}$$

♦ *Mean mark 35%.*

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26. Financial Maths, STD2 F4 2015 HSC 29b

Without the lump sum payment

$$\begin{aligned}\text{Total repayments } &= 30 \times 12 \times \$1880\\ &= \$676\ 800\end{aligned}$$

With the lump sum payment

$$\begin{aligned}\text{Total repayments } &= (22 \times 12 \times \$1880) + \$80\ 000\\ &= \$496\ 320 + \$80\ 000\\ &= \$576\ 320\end{aligned}$$

$\therefore$ Amount Jamal saved
$= 676\ 800 - 576\ 230$
$= \$100\ 480$

♦ *Mean mark 34%.*

---

27. Financial Maths, STD2 F4 2017 HSC 28c

$\text{Interest per month } = \dfrac{12\%}{12} = 1\%$

Let $A =$ amount owing

After 1st repayment:

$$\begin{aligned}A_1 &= (100\ 000 + 1\% \times 100\ 000) - 1029\\ &= \$99\ 971\end{aligned}$$

After 2nd repayment:

$$\begin{aligned}A_2 &= (99\ 971 + 1\% \times 99\ 971) - 1029\\ &= \$99\ 941.71\end{aligned}$$

♦ *Mean mark 41%.*

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28. Financial Maths, STD2 F4 2018* HSC 28d

i. $\text{Days of interest } (n) = 24$

$\text{Daily interest rate } (r) = \dfrac{0.1241}{365} = 0.00034$

$$\begin{aligned}\text{Balance Owing } (FV) &= PV(1+r)^n\\ &= 849(1.00034)^{24}\\ &= 855.954\ldots\\ &= \$855.95 \ \ \text{(nearest cent)}\end{aligned}$$

ii. $$\begin{aligned}\text{Balance owing } &= 855.95 - 450 + 3\\ &= \$408.95\end{aligned}$$

♦ *Mean mark part (i) 48%, part (ii) 49%.*

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29. Financial Maths, STD2 F4 2019 HSC 27

$$\begin{aligned}\text{Daily interest } &= \dfrac{18.25}{100 \times 365}\\ &= 0.0005\end{aligned}$$

$$\begin{aligned}\text{Closing balance } &= 3700(1.0005)^{11}\\ &= 3720.40\end{aligned}$$

$$\begin{aligned}\therefore \text{ Minimum payment } &= 3720.40 \times 0.02\\ &= \$74.408\ldots\\ &= \$74.41 \ \ \text{(nearest cent)}\end{aligned}$$

♦ *Mean mark 39%.*

30. Financial Maths, STD2 F4 2023 HSC 32

a. Daily interest rate $(r) = \dfrac{13.5}{365}\% = \dfrac{0.135}{365}$

♦ Mean mark (a) 45%.

$n =$ 21 days

$$FV = PV(1+r)^n$$
$$= 450\left(1+\frac{0.135}{365}\right)^{21}$$
$$= \$453.51$$

$\therefore$ Interest charged $= 453.51 - 450 = \$3.51$

b. Interest as % total repaid $= \dfrac{3.51}{453.51} \times 100$

♦ Mean mark (b) 49%.

$$= 0.007739\ldots$$
$$= 0.77\,\% \text{ (to 2 d.p.)}$$

31. Financial Maths, STD2 F4 2022 HSC 36

a. Monthly interest rate $= \dfrac{7.2}{12} = 0.6\%$

$$A = 199\ 715 \times \frac{0.6}{100}$$
$$= \$1198.29$$

$$B = P + I - R$$
$$= 199\ 428.29 + 1196.57 - 1485$$
$$= \$199\ 139.86$$

b. Total payments if lump sum not paid

♦♦♦ Mean mark (b) 17%.

$$= (23 \times 12) \times 1485$$
$$= \$409\ 860$$

Total payments if lump sum paid

$$= 40\ 000 + (50 + 146) \times 1485$$
$$= \$331\ 060$$

Savings by paying the lump sum

$$= 409\ 860 - 331\ 060$$
$$= \$78\ 800$$

32. Financial Maths, STD2 F4 2023 HSC 29

a. 8.0% interest over a 25 year loan

♦ Mean mark (a) 50%.

Monthly repayments to borrow $1000 = $7.72

Total months $= 25 \times 12 = 300$

Monthly repayments $= 520 \times 7.72$

$$= \$4014.40$$

$\therefore$ Total repayments $= 4014.40 \times 300$

$$= \$1\ 204\ 320$$

♦♦♦ Mean mark (b) 11%.

b. 7.0% interest over a 15 year loan

Monthly repayments to borrow $1000 = $8.99

$\therefore$ Amount borrowed $= \dfrac{3596}{8.99} \times \$1000$

$$= 400\ 000$$